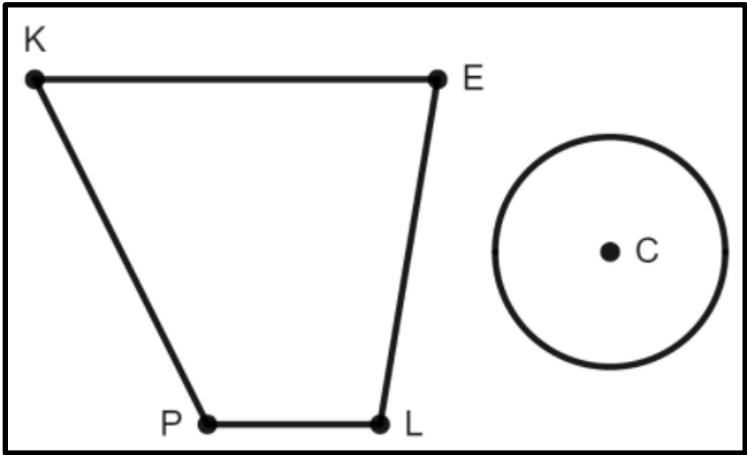


A diagram with Trapezoid *KELP* and Circle *C* is shown. Given the dimensions, find the area of each figure and complete the table. Round to the nearest hundredth if necessary.



Shape	Dimensions	Area
Trapezoid <i>KELP</i>	$h = 14.2, b_1 = 13.7, b_2 = 5.6$	1. $\frac{1}{2}(13.7 + 5.6)(14.2)$ $= 137.03 \text{ units}^2$
Circle <i>C</i>	$r = 9.8$	2. $\pi(9.8)^2$ $\approx 301.72 \text{ units}^2$ (when using the pi button)

Determine the area of the dilated figures and complete the table.

Shape	Area with Scale Factor of $\frac{5}{2}$	Area with Scale Factor of $\frac{2}{5}$
Trapezoid <i>K'E'L'P'</i>	3. $(137.03)\left(\frac{25}{4}\right) = 856.44 \text{ units}^2$	4. $(137.03)\left(\frac{4}{25}\right) \approx 21.92 \text{ units}^2$
Circle <i>C'</i>	5. $(301.72)\left(\frac{25}{4}\right) = 1888.75 \text{ units}^2$	6. $(301.72)\left(\frac{4}{25}\right) \approx 48.28 \text{ units}^2$

7. Parallelogram *KEYS* has a base of 8.5 feet and a height of 12.5 feet. What is the area of Parallelogram *KEYS*?

$$A = bh = (8.5)(12.5) = 106.25 \text{ ft}^2$$

8. Parallelogram *KEYS* is multiplied by a scale factor to create Parallelogram *K'E'Y'S'*. Parallelogram *K'E'Y'S'* has an area of 425 square feet. What is the scale factor for the area of the pre-image to get the image?

$$\frac{425}{106.25} = 4 \text{ area ratio}$$

$$\text{Scale factor of dimensions} = 2$$

9. Using the scale factor, what is the base of the image?

$$8.5 \times 2 = 17 \text{ ft}$$

10. Using the scale factor, what is the height of the image?

$$12.5 \times 2 = 25 \text{ ft}$$